Clinical Natural Language Technology

for Health Care: Past, Present & Future

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Cotiviti GenAI Science Internship · Demonstration

github.com/akhilg-9/cotiviti-cni-assessment

The Problem

Why clinical language technology matters to Cotiviti

~80%

of healthcare data is unstructured text

$31.7B

Medicare FFS improper payments, FY2024 (CMS, 7.66%)

- Notes, discharge summaries, path/radiology reports, scanned charts.
- That text hides coding accuracy, quality gaps, recoverable dollars.
- Reading it at scale is a direct lever on Treatment, Payment & Operations.

Past → Present → Future

Three eras of clinical NLP

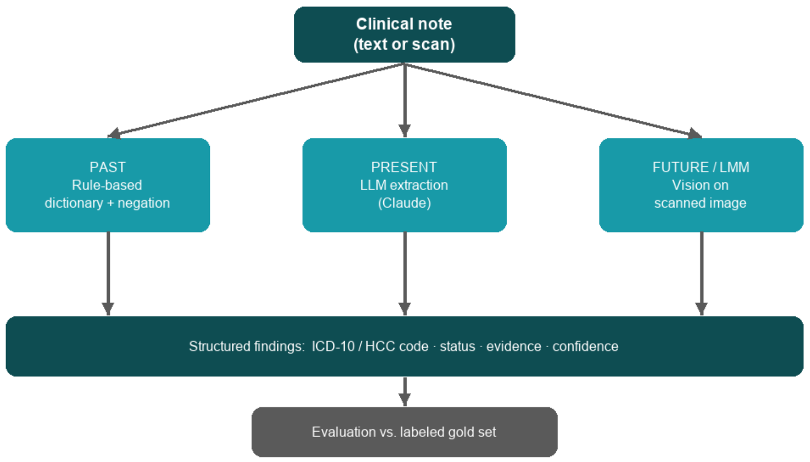
- PAST — rule-based: dictionaries, regex, ontologies (UMLS/SNOMED/ICD), NegEx.
  - Transparent and deterministic, but brittle: finds only what it was told about.
- PRESENT — LLM structured extraction: reads the note in context.
  - Normalizes phrasing, codes to ICD-10, reasons about negation and uncertainty.
- FUTURE — multimodal (LMM): the model reads the scanned page image directly.
  - The separate OCR stage disappears.

Proof of Concept

One pipeline, three eras, same note — measurable

- Free-text note → codeable ICD-10 / HCC findings.
- Every finding carries status, verbatim evidence, and confidence.
- Schema-enforced JSON — malformed model output is rejected.
- Versioned prompts, 17 unit tests, ruff-linted CI, Streamlit UI.
- Validated live on real de-identified notes (discharge summaries, an H&P, five MTSamples transcriptions).

Clinical Note Intelligence — Architecture



Measured Results — Labeled Gold Set

cni eval · 9 conditions across 2 notes · live claude-haiku-4-5

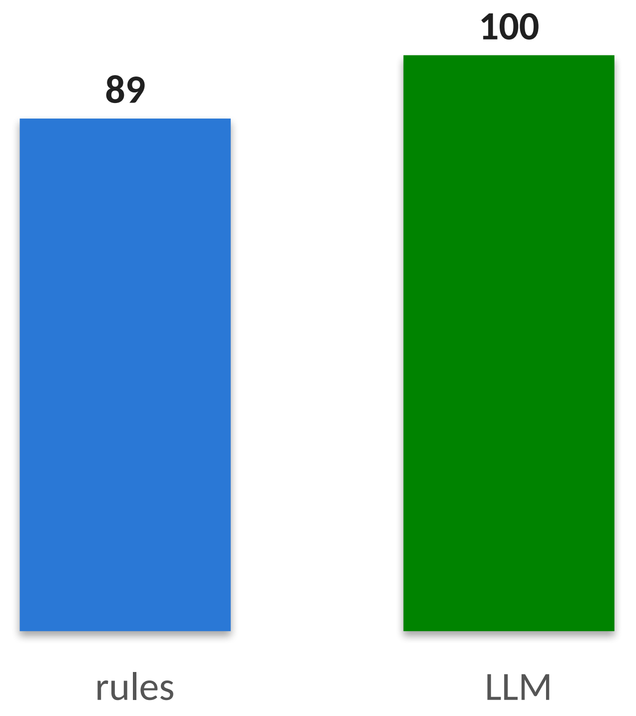
Era	Precision	Recall	F1	Status accuracy
Past — rules	1.00	1.00	1.00	89%
Present — LLM	1.00	1.00	1.00	100%

The note says hypertension is "suspected, not yet confirmed."

Rules code it ACTIVE (backward-only negation scan). The LLM reads

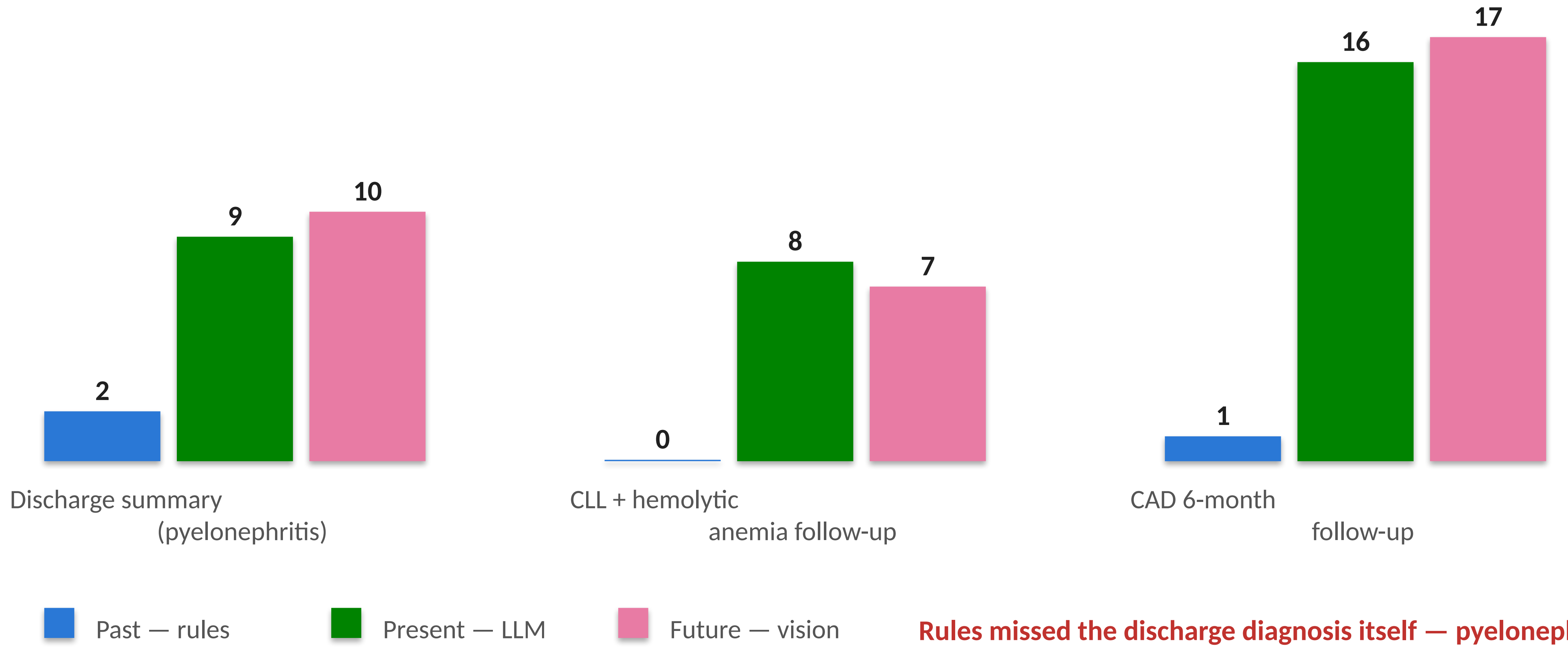
the sentence and says UNCERTAIN — the 89% → 100% gap is exactly that.

Status accuracy — where the eras separate



Real Notes, Real Gap

Findings per era on real de-identified notes (live runs)



Rules missed the discharge diagnosis itself — pyelonephritis (N10).

The Failure That Matters

Found organically on a real cardiology note

"...cardiac catheterization showed NO EVIDENCE OF coronary artery disease."

- Rule-based era: reports CAD as ACTIVE — the negation pattern never fires.
- LLM era: correctly omits it.
- That is a false diagnosis a payer would have had to defend.
- Pinned as a unit test — the brittleness is documented behavior, not an anecdote.

Guardrails Built In

A deterministic guard + audit trail on a probabilistic model

74,719	97.8%	92.1%
official ICD-10-CM codes every emitted code checked	LLM codes valid across real-note runs	vision codes valid every miss flagged

- Flagged live: R61.9, R21.9 — well-formed codes that don't exist.
- Code-set drift caught: M54.5, D59.1, C91.1 were valid in earlier fiscal
  - years, since subdivided — the model codes from its training era; the FY2026 table catches it.
- Every run writes an audit artifact: note, model, prompt version, findings + evidence.

Opportunities & Threats

Opportunities

• Chart-to-code automation for risk adjustment (HCC) and payment integrity.

• Policy → executable rules; summarize and compare policy versions.

• Quality-measure abstraction (HEDIS) from free text.

• Clinical validation for prior authorization and appeals.

Threats

• Hallucinated codes/evidence — unacceptable when payment depends on it.

• PHI / HIPAA and data-governance obligations.

• Regulatory scrutiny (FDA, ONC); auditability demands.

• Model bias/equity; vendor cost and lock-in.

Recommended Actions

Differentiate on trustworthy, explainable clinical AI

- Governed, human-in-the-loop chart review with span-level evidence per code.
- Pair LLM extraction with deterministic ICD-10/HCC validation (as this POC does).
- Adopt multimodal models for scanned/faxed records — retire brittle OCR pipelines.
- Stand up evaluation + guardrail frameworks before production.

Key Takeaways

1

The eras separate on judgment, not detection

Status accuracy 89% → 100%; on real notes the dictionary missed the discharge diagnosis itself.

2

Trust is an architecture, not a model property

Official code-table validation + evidence spans + audit artifacts make LLM output defensible.

3

This maps directly to Cotiviti's franchise

Chart-to-code for risk adjustment and payment integrity, with $31.7B/yr of improper payments in scope.

Thank you

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